Evaluating Domain Adaptation Techniques for Classical Machine Learning Models

Research report for Domain Shift Workbench

The completed study found that adaptation was useful in some transfers and harmful in many others. Of 132 primary task-model-adaptation comparisons, mean macro F1 increased in 40 and decreased in 92. CORAL helped gas classification more consistently than it helped HAR or Amazon sentiment. These results support conditional use of adaptation, with a matched source-only baseline and visible failure diagnostics. They do not establish a universally superior method.

This report covers the frozen protocol, implementation, all completed comparisons, their limitations and reproducibility. Student registration, institution, supervisor and approval details were not supplied and are omitted. The separate assistance disclosure describes the implementation support actually used.

Abstract

Distribution shift can reduce the reliability of classical classifiers. This study evaluates domain-classifier importance weighting and centred, regularised CORAL against source-only Logistic Regression, linear Support Vector Machine and Random Forest models. Experiments use UCI Gas Sensor Array Drift, smartphone Human Activity Recognition and the original four-domain Amazon sentiment corpus, covering 22 transfers and five seeds. Source-only validation selects shared hyperparameters; separate target adaptation features support adaptation, while held-out target observations remain excluded until evaluation. The completed study contains 1,100 primary and 180 sensitivity evaluations. Across 132 primary adaptation comparisons, mean macro F1 improves in 40 and declines in 92; 27 descriptive group-bootstrap intervals include zero. CORAL benefits several gas transfers but adaptation generally harms HAR and Amazon performance. Accuracy, runtime, weight concentration and covariance diagnostics contextualise these outcomes. Shared test observations and few independent domains limit inference. The contribution is reproducible comparative evidence and an independently auditable local research workbench.

1 Introduction and objective

A classifier can encounter different input distributions after deployment. Gas sensors change across acquisition periods, human activity features vary between people, and expressions of sentiment vary between product categories. Here a domain is a source or target setting within one dataset. No classifier combines unrelated gas, activity and sentiment features into one training problem.

The exact assessment objective is:

To evaluate whether importance weighting and feature alignment using CORAL improve the performance of Logistic Regression, Support Vector Machine, and Random Forest when source and target data follow different distributions. The study will compare these techniques against models trained without adaptation across three public datasets, using accuracy, macro F1 score, and computational time. It will identify the conditions under which adaptation improves performance, provides little benefit, or reduces prediction quality.

Three research questions organize the analysis. First, how do importance weighting and CORAL change target macro F1, accuracy and computational time against the matched source-only model? Second, how do effects differ across temporal gas transfers, disjoint HAR participants and Amazon categories? Third, what do weight concentration and covariance diagnostics reveal about useful, uncertain and harmful adaptation? The contribution is a reproducible empirical study and an inspectable application. The methods themselves are established work.

2 Related work

Covariate-shift weighting assumes that input frequencies change while the conditional label relationship remains suitable for transfer. Sugiyama et al. develop importance-weighted validation [R02]; Huang et al. estimate weights by kernel mean matching [R03]. Bickel et al. show how discrimination between training and test inputs relates to density ratios [R11]. This project implements a two-stage domain-classifier weighting procedure. It does not implement IWCV, kernel mean matching or an integrated discriminative optimizer.

CORAL matches second-order statistics [R01]. Fernando et al. instead align eigenvector subspaces [R06]. Matching a covariance matrix is different from matching class-conditional distributions. Theoretical domain-adaptation bounds include the feasibility of predicting well in both domains [R04], helping explain why small marginal discrepancy alone is insufficient. Pan and Yang provide the terminology and discuss negative transfer [R07].

The Amazon corpus and structural correspondence learning were introduced in the sentiment study of Blitzer et al. [R05]. Vergara et al. motivate the gas-drift setting with supervised temporal ensembles [R08]. Anguita et al. document the HAR representation and overlapping windows [R09]. Those published protocols differ from this one, so their scores are not treated as direct baselines. The scikit-learn paper [R10] supplies the software citation; official version-matched documentation [S01-S06] supports the actual interfaces. The companion literature matrix compares eleven papers and records restricted full-text access explicitly.

3 Data and frozen partitions

The original archives were acquired on 10 October 2026. Parser version 2.0.0 retained the published numerical feature dimensions. Numerical inputs must be finite; unexpected missing values fail validation rather than triggering unreported imputation. Exact feature values, not rounded features, define numerical duplicate groups. All archive hashes, acquisition timestamps, class maps, exclusions and source label counts are recorded in the completed data cards.

Dataset	Parsed	Retained	Excluded	Fields	Tasks
gas	13910	13910	0	128	9
har	10299	10299	0	561	1
amazon	78679	70509	8170	1	12

Gas uses Batch 1 as the source for nine later batches. A fixed duplicate-safe partition divides each target batch approximately in half without consulting its labels. Batch 4 has 80 adaptation and 81 evaluation examples; this is a within-batch holdout, not fine-grained chronological forecasting. All 13,910 observations were retained. Gas uses 128 features and six canonical classes [D01].

HAR uses 7,352 windows from 21 official training participants as source. Target adaptation contains 1,201 windows from participants 2, 4, 9 and 10; evaluation contains 1,746 windows from participants 12, 13, 18, 20 and 24. All 10,299 windows were retained. Source validation and the target roles keep whole people separate. The test concerns adaptation to a population of different people, not calibration to the same individual. The 561 features and six activities are the official representation [D02].

Amazon uses books, DVDs, electronics and kitchen as four domains within one dataset. All twelve ordered source-target pairs are evaluated. Of 78,679 parsed records, 8,170 duplicate bodies were excluded and 70,509 remained. Retained labeled counts are 1,956 books, 1,954 DVDs, 1,965 electronics and 1,977 kitchen reviews. The corresponding unlabeled pools contain 5,473, 29,281, 12,405 and 15,498 bodies. One additional source-target

duplicate is excluded from DVD adaptation in the books-to-DVD transfer. The separate original books unlabeled file was included. Only review bodies enter features; ratings, file-label fields, titles and reviewer metadata do not. Product identifiers are retained separately for grouping [D03].

Split seed 2026 fixes every role ID before model fitting. The same evaluation rows are used for all methods and five model seeds. The complete 22-task count table and split hashes accompany this report. No raw archive is included in the source package: the gas landing page has both research-only wording and a CC BY 4.0 section; HAR states CC BY 4.0; the Amazon page has no explicit redistribution license. Acquisition instructions and citations are provided instead.

4 Methods and fair comparison

Logistic Regression (LR), LinearSVC and Random Forest (RF) are fitted separately on each dataset. The source-only model receives source features and labels. The source-majority baseline always predicts the most frequent source class, with deterministic tie breaking. Source-selected estimator parameters are shared between source-only, importance weighting and CORAL within each paired comparison.

Common representation and source selection

Gas and HAR use source-fitted standard scaling. Amazon uses lowercase unigram and bigram TF-IDF with min_df 2, at most 20,000 features, sublinear term frequency and no default stopword list. Source-fitted TruncatedSVD reduces this to at most 128 components below available rank, followed by source-fitted scaling. The same dense representation is used by every primary method. Preprocessing is fitted again inside each source validation fold, excluding that fold's validation rows.

Five-fold source macro F1 selects C from 0.1, 1 and 10 for LR and LinearSVC. RF uses 200 trees and the four combinations of maximum depth unlimited or 20 and minimum leaf size 1 or 3. HAR folds preserve participant groups; Amazon uses product groups when feasible. Thirty shared source searches produced 1,500 candidate-fold scores. Source-only selection keeps the comparison controlled, but it need not choose the target-optimal model under distribution shift.

Importance weighting

A separate logistic classifier predicts domain membership: source 0 or target adaptation 1. It uses C 1, without class balancing or resampling. Five source folds yield out-of-fold domain probabilities, training each discriminator on the other source folds and all permitted target adaptation features. True target task labels are not accepted by the training interface.

For a source observation, let q be the estimated target-domain probability, ns the actual source count in its discriminator training fold, and nt the target adaptation count. The implementation bounds q to [0.001, 0.999], calculates `r = q / (1 - q) * ns / nt`, clips r to [0.05, 10], then divides by the global mean of the clipped source weights. This prior correction removes the domain sampling imbalance; final normalized weights need not remain within the raw clipping interval. Each task estimator receives these weights through sample_weight.

Effective sample size is `ESS = sum(w)^2 / sum(w^2)`. ESS below 10% of the source count produces a retained diagnostic warning. Weighting cannot create labeled source support in a region absent from the source. Incorrect probabilities or a changed conditional label relationship can make the weighting harmful. The density-ratio motivation is related to R11; the fold and clipping choices are the frozen implementation contract of this study.

Centred regularised CORAL

For row-vector source and target-adaptation features, compute means ms and mt, and unbiased covariances Cs0 and Ct0. Set `Cs = Cs0 + 0.001 I` and `Ct = Ct0 + 0.001 I`. Symmetric eigendecomposition with eigenvalue floor 1e-8 gives `A = Cs^(-1/2) Ct^(1/2)`. Train on `(Xs - ms) A + mt` and source labels. Predict target observations after the common source preprocessing; do not apply the source alignment matrix to target inputs. This centred variant with mean restoration is explicitly distinguished from the original CORAL presentation [R01].

The recorded mismatch is the Frobenius covariance difference divided by the target covariance norm, floored at 1e-12 in the denominator. Ridge regularization means the empirical covariances need not match exactly. Reduced mismatch is a geometric diagnostic, not proof that the task decision boundary improved.

5 Implementation and validation

The Python package contains data contracts, parsers, preprocessing, source selection, adaptation, fitting, evaluation and paired analysis. The CLI and Streamlit workbench share the engine. Six pages expose actual saved evidence. An explicit worker handles jobs, ownership, logs and cancellation; rerenders do not train a model. The repository reader validates seals and file hashes. Each displayed comparison links to run IDs, the frozen split and prediction checksums.

The data flow is: official archive to immutable parsed data to frozen split; source folds to shared selection and preprocessing; source labels plus target adaptation features to model fitting; completed-fit authorization to the evaluator; saved predictions to metrics and group-bootstrap analysis; saved tables to UI and exports. Held-out features and labels are excluded from fitting, including domain discrimination, covariance estimation and tuning.

The tested machine has an Intel i7-14700HX, 16 GB RAM and RTX 4060 8 GB. Computation uses one CPU job and one numerical thread; the GPU is unused. CPython 3.12.10, NumPy 2.5.3, SciPy 1.18.1, pandas 2.3.3, scikit-learn 1.9.1 and Streamlit 1.65.0 are among 64 actual locked dependencies. The frozen scientific snapshot contains 38 files. Source, archive, split, configuration and dependency identities are recorded before final evaluation.

Numerical and acceptance tests cover prior correction with unequal domain counts, independent covariance checks, group separation, fold-only preprocessing, held-out read refusal, sample-weight delivery, metric fixtures, mutation detection, atomic artifacts and resume. The completed dashboard regression gate passed 212 tests, followed by 20 final interface checks; the current delivery audit adds report/package checks. Six pages were exercised with real and empty storage; 16 actual Chrome screenshots cover desktop, tablet and detailed views. The verification receipts distinguish automated checks from student understanding.

6 Evaluation and results

The completed primary matrix contains 990 classifier evaluations and 110 majority evaluations. Predeclared sensitivity adds 180 evaluations, for 1,280 total. There are no missing comparisons or classifier fitting failures. Seeds are 17, 29, 43, 71 and 101. Metrics are recomputed from persisted predictions rather than edited summary values. Macro F1 averages classes present in the full held-out task, with zero_division 0; absent classes are recorded. Accuracy, macro precision/recall, per-class support and confusion matrices accompany it.

Each paired delta compares the same ordered evaluation IDs, seed, preprocessing recipe, selected base parameters and dataset/split identities. The reported uncertainty is a descriptive 95% paired percentile interval from 1,000 group resamples with bootstrap seed 2026. Each resample averages the five fitted-seed deltas without refitting. Gas uses duplicate-safe measurement groups, HAR whole evaluation participants, and Amazon product groups

where available. The intervals are conditional on the fitted models and observed domains and are unadjusted for multiple comparisons. Five seeds are not independent datasets.

Dataset summaries

The following table uses equal task weights within each dataset, after averaging the five seeds for each task. Scores are fractions and time is seconds. These dataset averages do not pool unrelated observations or create an independent sample size. Per-task mean/SD, paired intervals and exact run IDs remain in the accompanying full tables.

GAS

Model	Method	Accuracy	Macro F1	Pipeline s	F1 delta pp
Linear SVM	CORAL	0.596098	0.555106	0.123	+11.348
Linear SVM	Weighting	0.478365	0.455392	0.454	+1.376
Linear SVM	Source	0.463566	0.441627	0.094	-
LR	CORAL	0.610142	0.554631	0.250	+22.576
LR	Weighting	0.458709	0.312692	0.185	-1.617
LR	Source	0.450858	0.328867	0.080	-
Majority	Source	0.199823	0.058899	0.037	-
RF	CORAL	0.564881	0.467566	0.563	+4.791
RF	Weighting	0.426469	0.322044	0.498	-9.761
RF	Source	0.480556	0.419659	0.462	-

HAR

Model	Method	Accuracy	Macro F1	Pipeline s	F1 delta pp
Linear SVM	CORAL	0.933562	0.932259	6.752	-5.235
Linear SVM	Weighting	0.975372	0.975521	34.704	-0.909
Linear SVM	Source	0.984536	0.984613	7.403	-
LR	CORAL	0.942153	0.939980	5.754	-2.965
LR	Weighting	0.967354	0.966716	11.639	-0.291
LR	Source	0.970790	0.969630	4.639	-
Majority	Source	0.187285	0.052581	2.355	-
RF	CORAL	0.935510	0.928315	30.938	-1.285
RF	Weighting	0.936770	0.930070	22.753	-1.109
RF	Source	0.946850	0.941164	27.042	-

AMAZON

Model	Method	Accuracy	Macro F1	Pipeline s	F1 delta pp
Linear SVM	CORAL	0.693304	0.689062	10.719	-4.382
Linear SVM	Weighting	0.707230	0.706099	11.166	-2.678
Linear SVM	Source	0.733862	0.732884	6.134	-
LR	CORAL	0.699145	0.694939	10.695	-3.883
LR	Weighting	0.707991	0.706888	11.132	-2.688
LR	Source	0.734645	0.733769	6.117	-
Majority	Source	0.505741	0.335873	6.127	-
RF	CORAL	0.694431	0.690214	14.369	+0.128
RF	Weighting	0.659609	0.646201	12.774	-4.273
RF	Source	0.691885	0.688930	9.668	-

CORAL's average gas gains are concentrated in LR and linear SVM; the method is not beneficial for every gas transfer. The HAR source-only models already perform strongly, and all six primary adaptation comparisons have lower mean macro F1. For Amazon, both adaptations usually reduce the linear models' macro F1. RF with CORAL has a small positive average across Amazon tasks despite mixed task effects. An average should therefore be read alongside the direction counts and task plots.

Dataset	Method	Pairs	Positive	Negative	CI includes 0
amazon	CORAL	36	8	28	7
amazon	Weighting	36	3	33	4
gas	CORAL	27	19	8	5
gas	Weighting	27	10	17	9
har	CORAL	3	0	3	1
har	Weighting	3	0	3	1

Across all 132 primary comparisons, 40 means increase, 92 decrease and none are exactly unchanged. Twenty-seven macro-F1 intervals include zero. An interval including zero is uncertainty, not proof that two methods are equivalent.

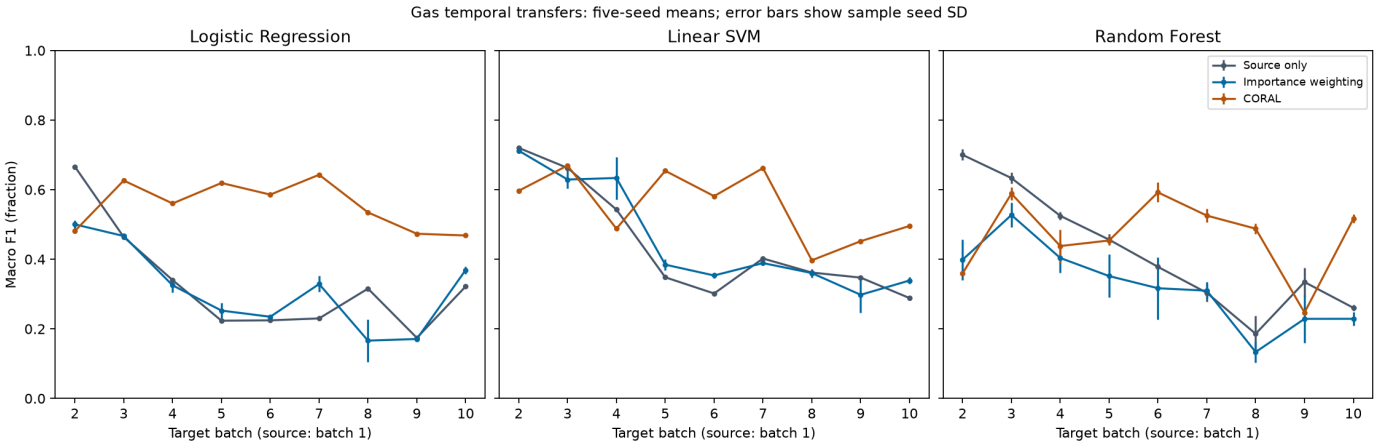


Figure 1. Gas Batch 1 to Batches 2-10, primary macro F1 in fractions, five-seed means with error bars showing sample seed SD. Batch identities and each trace's exact run IDs are in the original figure-trace manifest. Tasks share a source and are not independent replications.

Positive uncertain and negative examples

Example	Transfer	Model	Method	Delta pp	95% interval pp
positive	Gas 1→07	LR	CORAL	+41.284465	[+38.973957, +43.482547]
uncertain	Gas 1→08	Linear SVM	Weighting	-0.148662	[-1.266615, +0.828300]
negative	Gas 1→02	RF	CORAL	-34.075518	[-36.618668, -18.709011]

The positive example is CORAL with LR on gas Batch 7: mean improvement is 41.284465 percentage points. On gas Batch 8, importance weighting with linear SVM changes macro F1 by -0.148662 points, with an interval spanning zero. On gas Batch 2, CORAL with RF reduces macro F1 by 34.075518 points. These are descriptive examples selected for interpretation after the complete analysis; they did not select a new primary setting. The full task tables retain the other comparisons.

Variation and computational cost

gas_b01_b10

Model	Method	Accuracy mean ± SD	Macro F1 mean ± SD	Pipeline s
Linear SVM	CORAL	0.533333 ± 0.000000	0.495735 ± 0.000000	0.118
Linear SVM	Weighting	0.397333 ± 0.010149	0.338931 ± 0.010036	0.290
Linear SVM	Source	0.348333 ± 0.000000	0.288451 ± 0.000000	0.100
LR	CORAL	0.500556 ± 0.000000	0.468356 ± 0.000000	0.172
LR	Weighting	0.423111 ± 0.012175	0.369173 ± 0.010679	0.202
LR	Source	0.369444 ± 0.000000	0.321932 ± 0.000000	0.079
Majority	Source	0.172778 ± 0.000000	0.049108 ± 0.000000	0.042
RF	CORAL	0.568444 ± 0.011093	0.516200 ± 0.012263	0.595
RF	Weighting	0.313222 ± 0.012487	0.228830 ± 0.019438	0.535
RF	Source	0.333889 ± 0.007233	0.260502 ± 0.008091	0.483

har_official_group_transfer

Model	Method	Accuracy mean ± SD	Macro F1 mean ± SD	Pipeline s
Linear SVM	CORAL	0.933562 ± 0.000000	0.932259 ± 0.000000	6.752
Linear SVM	Weighting	0.975372 ± 0.000000	0.975521 ± 0.000000	34.704
Linear SVM	Source	0.984536 ± 0.000000	0.984613 ± 0.000000	7.403
LR	CORAL	0.942153 ± 0.000000	0.939980 ± 0.000000	5.754
LR	Weighting	0.967354 ± 0.000000	0.966716 ± 0.000000	11.639
LR	Source	0.970790 ± 0.000000	0.969630 ± 0.000000	4.639
Majority	Source	0.187285 ± 0.000000	0.052581 ± 0.000000	2.355
RF	CORAL	0.935510 ± 0.002483	0.928315 ± 0.002614	30.938
RF	Weighting	0.936770 ± 0.001188	0.930070 ± 0.001163	22.753
RF	Source	0.946850 ± 0.001689	0.941164 ± 0.001827	27.042

amazon_books_electronics

Model	Method	Accuracy mean ± SD	Macro F1 mean ± SD	Pipeline s
Linear SVM	CORAL	0.657608 ± 0.006521	0.655290 ± 0.006620	7.613
Linear SVM	Weighting	0.665344 ± 0.006983	0.663719 ± 0.006617	7.848
Linear SVM	Source	0.704733 ± 0.003325	0.701793 ± 0.003351	6.068
LR	CORAL	0.664122 ± 0.006075	0.661715 ± 0.006342	7.584
LR	Weighting	0.664733 ± 0.006822	0.662942 ± 0.006390	7.837
LR	Source	0.705547 ± 0.005147	0.702958 ± 0.005129	6.063
Majority	Source	0.502290 ± 0.000000	0.334350 ± 0.000000	6.075
RF	CORAL	0.673893 ± 0.003440	0.670395 ± 0.003180	9.710
RF	Weighting	0.627684 ± 0.016657	0.624856 ± 0.019394	8.651
RF	Source	0.656896 ± 0.006164	0.655878 ± 0.006095	8.241

Table values above are saved five-seed means with sample SD, with primary settings only. A displayed SD of 0.000000 can represent deterministic predictions or floating-point rounding near zero; it does not remove test-sample uncertainty. Pipeline time sums original preprocessing fit, adaptation fit, classifier fit and target prediction time. Shared source tuning is reported separately, and the originating component time is retained for cached runs. Summing every row would double-count shared computation.

Dataset	Preprocess s	CV s	10 fits s	Wall s	Peak MiB
gas	0.017	20.403	1.575	25.361	284.2
har	3.497	749.701	145.266	943.342	687.0
amazon	3.845	50.542	4.957	68.054	517.0

The representative gas, HAR and Amazon benchmarks measured complete source search and ten primary fits at seed 17. Peak RAM is process-lifetime working set, including imports and persistence. Their study-wide planning projection was 2.98-3.97 hours from scratch, including stated allowance for unmeasured work. This is an estimate, not observed total elapsed completion time. The successful resumed invocation lasted 2,162.844113 seconds and the first completed fitting boundary recorded 7,629.014105 seconds; these scopes overlap and must not be added. Two Windows progress-file replacement interruptions were retained and resumed. They were operational failures, while classifier fitting failures were zero.

Diagnostics and sensitivity

The saved diagnostics retain 180 low-ESS warnings across the completed fit records. These are not 180 independent failures: transformations are shared across models. Concentrated weights indicate limited effective source support, but cannot alone identify the causal reason for negative transfer. CORAL can greatly reduce the covariance residual while worsening macro F1, as seen in the saved gas comparison.

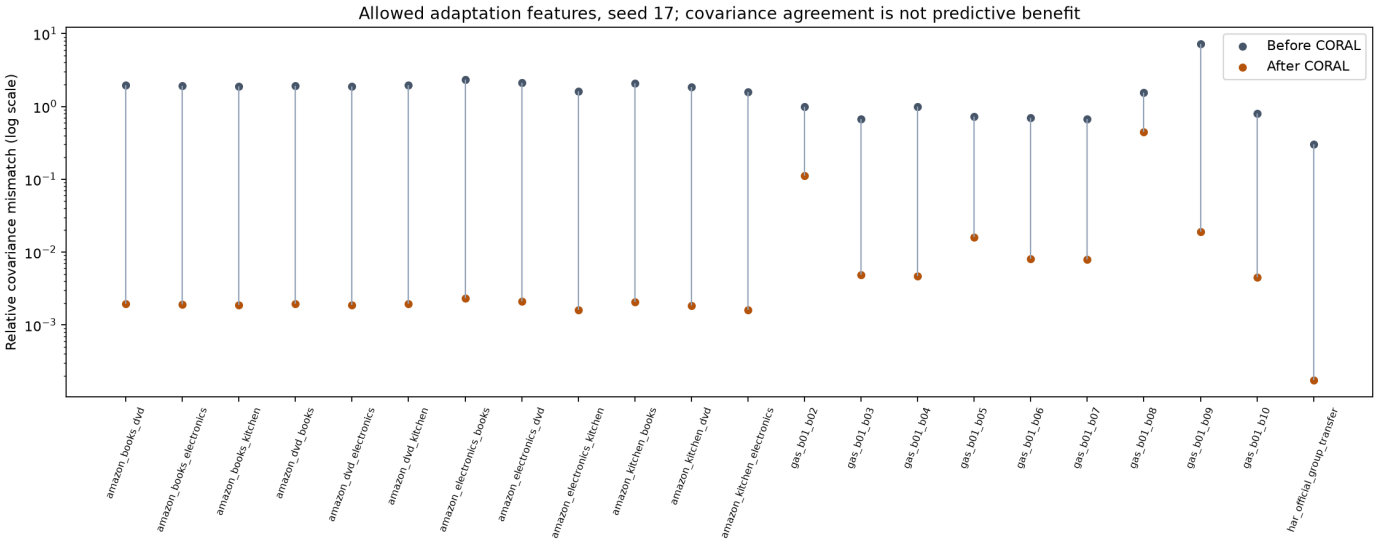


Figure 2. Saved CORAL covariance residuals before and after alignment for the declared diagnostic tasks. Residuals are normalized Frobenius quantities, not classification scores; the plotted positive values use a logarithmic display. The figure manifest links their exact input diagnostics and run IDs.

Clipping bounds 5, 10 and 20 and ridge values 0.0001, 0.001 and 0.01 were evaluated on gas Batch 10, HAR and books-to-electronics with all five seeds and three models. For gas linear SVM, increasing clipping from 5 to 20 changes mean macro F1 from 0.319973 to 0.354481, while the primary value 10 gives 0.338931. For HAR RF the corresponding values fall from 0.934667 to 0.923518. These different responses argue against choosing one new clipping bound from held-out performance.

For gas LR, ridge 0.0001, 0.001 and 0.01 gives mean macro F1 0.441869, 0.468356 and 0.494883. For books-to-electronics LR it gives 0.661510, 0.661715 and 0.663333. All declared settings, SDs, deltas and run IDs remain available in the sensitivity tables. The primary ridge stays 0.001 and primary clipping stays 10.

7 Discussion and limitations

The first research question has a conditional answer: both gains and losses occur, and method choice depends on the dataset, classifier and transfer. The second question shows a clear contrast in this study between several useful gas alignments and mostly negative HAR/Amazon adaptation. For the third, ESS and covariance diagnostics make the behavior inspectable but do not prove mechanisms. Strong source performance can leave little room for improvement, and an unsupervised marginal adjustment may move useful class structure in the wrong direction. These are plausible interpretations, not a causal identification.

Source-only hyperparameter selection was fixed for fairness and may disadvantage a method that would benefit from a different target-aware selection criterion. The density-ratio model can be misspecified, and covariate-shift support and conditional-label assumptions are not established by these experiments. CORAL only uses first/second-order information in the specified dense representation. The 128-component text representation limits vocabulary directions available to all methods, including source-only.

Only three independent applications are included. Amazon tasks reuse categories and reviews, gas transfers share Batch 1, and HAR has five evaluation people. Group bootstrap reduces some dependence but does not make these populations independent or representative of all future deployments. Finer gas timestamps are unavailable for a complete within-batch dependence model. Target-present class sets differ between some gas batches. Unadjusted intervals and repeated seeds must not be converted into a universal significance claim.

The runtime projection is specific to the tested CPU and thread settings. Reproduction on the same laptop checks installation and deterministic behavior, not portability to every OS or hardware stack. Interface keyboard tests cover the recorded paths and do not certify screen-reader accessibility. Institutional formatting and student rehearsal remain human submission responsibilities.

8 Reproduction and conclusion

A clean checkout and fresh virtual environment reproduced gas Batch 1 to Batch 2 using all five seeds and 50 newly fitted models/baselines. All 50 ordered prediction vectors, metric dictionaries and selected parameters exactly matched the original study on 622 evaluation rows per run. Fitting took 204.295694 seconds; fitting, evaluation and comparison took 245.421423 seconds. Peak process working set was 216.0625 MiB. No forbidden held-out training reads occurred. The original archive/split were verified and reused; new caches/models/predictions were written separately. The detailed reproduction record preserves the exact executed driver and logs.

The reproducibility guide gives exact environment, acquisition, execution, checkpoint/resume and export commands. The package includes permitted code, configuration, aggregate results, screenshots and hashes. Exact local sample IDs and prediction files remain inspectable through the saved study; data acquisition instructions replace redistribution of raw archives. The supplied original pack MANIFEST.sha256 still describes its original files; new deliverables use a separate manifest rather than rewriting that historical record.

The completed evidence answers the locked objective without assuming adaptation must succeed. CORAL can improve classical models under some gas transfers, but neither covariance alignment nor importance weighting is uniformly reliable across these applications. A matched source-only baseline, negative-transfer analysis, explicit data roles and reproducible evidence are central to interpreting an adaptation result. Further work could study independent populations, target-data budgets or justified model-selection strategies under a new protocol.

Appendix A Original proposal abstract

The following planned-work abstract is preserved separately from the results abstract above. It contains the original 150 words.

Machine learning models can perform well on training data but struggle when applied to data collected under different conditions. This project will investigate whether domain adaptation can reduce this performance gap for Logistic Regression, Support Vector Machine, and Random Forest classifiers. It will compare importance weighting and correlation alignment (CORAL) with models trained without adaptation. Experiments will use the Gas Sensor Array Drift dataset, Human Activity Recognition Using Smartphones dataset, and Amazon Multi-Domain Sentiment dataset. These datasets represent changes over time, between users, and across product categories. Adaptation will use labelled source data and unlabelled target data, while target labels will be reserved for final evaluation. Accuracy, macro F1 score, and computational time will be measured across repeated experiments. The study aims to identify when adaptation improves predictions, when it offers limited benefit, and when it reduces performance, helping explain which approaches are useful under different types of distribution shift.

Appendix B Evidence and contribution record

The full per-task means/SD, paired seed changes, bootstrap intervals, per-run timings and sensitivity outputs are in the source package under reports/study. The companion completed data cards and split table record dataset provenance. The interface review links real screenshots to current app source hashes. The requirements checklist maps claims to executable checks and artifacts.

Codex provided substantial implementation, testing, execution, analysis tooling, writing and document-review assistance. The student's request fixed scope and selected the 150-word results abstract. No independent student implementation, rehearsal, faculty approval or publication acceptance is inferred from Git author metadata. The factual assistance record identifies what is observed and what the student must independently review and be able to defend.

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Checked against the primary publisher, proceedings, author or official software source on 11 October 2026. Method and dataset claims below are attributed to their original sources. Project scores come from the saved local study, not these papers.

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S04. Scikit-learn developers. StratifiedGroupKFold, version 1.9.1 documentation. API.

S05. Scikit-learn developers. TfidfVectorizer, version 1.9.1 documentation. API.

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Verification scope

R01-R05, R07, R09-R11 were checked using primary full-paper text, including the method or dataset sections used here. R06 was checked using the indexed CVF proceedings abstract and metadata; direct HTML/PDF access returned HTTP 403. R08 was checked using the indexed publisher abstract, introduction and method preview; direct publisher access also returned HTTP 403. No unseen experimental detail from either restricted paper is claimed. IEEE's page required JavaScript; the author-hosted R07 manuscript supplied the text. The DOI, volume and page metadata identify the final publication separately from that manuscript.

D01-D03 and S01-S06 were opened directly. The online scikit-learn documentation identified itself as 1.9.1, matching the installed scientific environment. The reading matrix records limitations and relevance; it does not claim that the student independently read or verified every source.